

per capita consumption of whole milk (x)
and of low-fat milk (y)

Per capita consumption (gallons)			
	1980	1984	1986
Whole milk (x)	17.1	14.7	12.8
Low-fat milk (y)	10.6	11.5	13.2

$$u = x + b$$

$$v = y + d$$

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \quad \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \quad \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$$

	i		
	1	2	3
u_i	4.3	1.9	0
v_i	0	0.9	2.6

$$\bar{u} = \frac{4.3 + 1.9 + 0}{3} = 2.0667$$

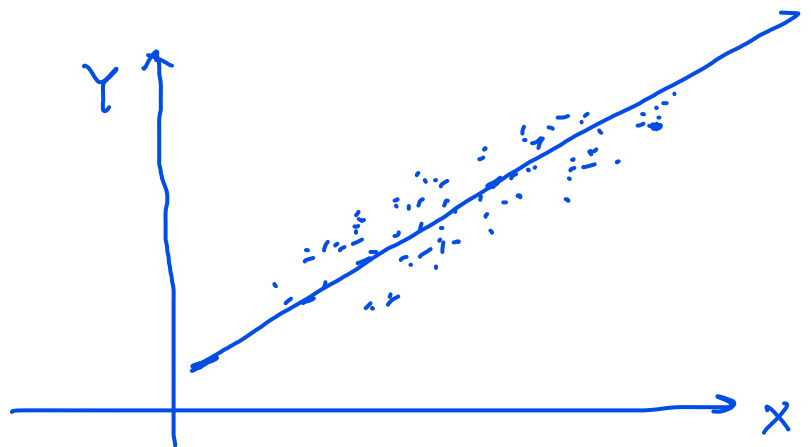
$$\bar{v} = \frac{0 + 0.9 + 2.6}{3} = 1.1667$$

$$\sum u_i v_i = (1.9)(0.9) = 1.71$$

$$\sum u_i^2 = (4.3)^2 + (1.9)^2 = 22.10$$

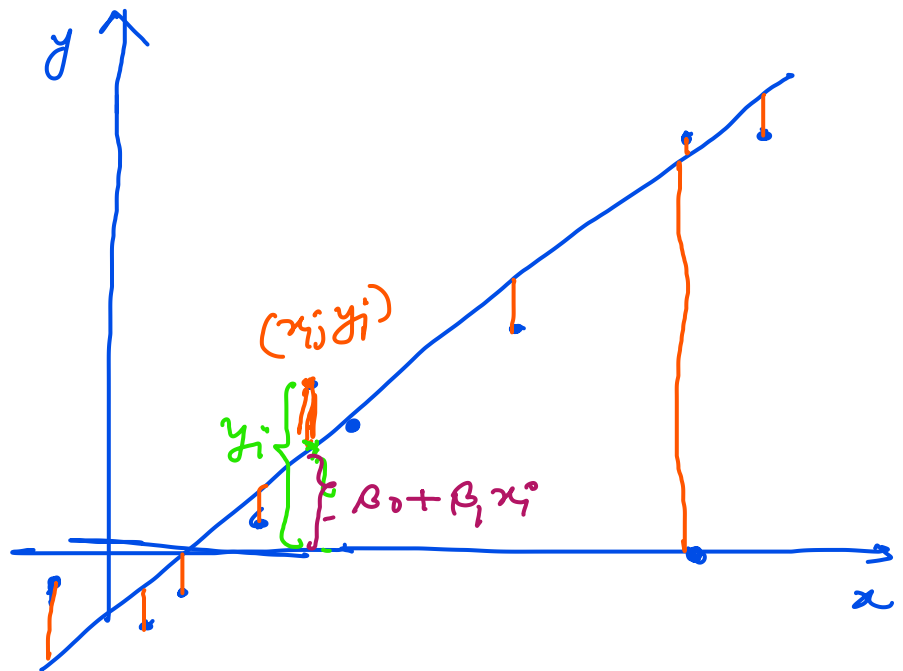
$$\sum v_i^2 = (0.9)^2 + (2.6)^2 = 7.57$$

$$r = \frac{1.71 - 3(2.0667)(1.1667)}{\sqrt{\{22.10 - 3(2.0667)^2\} \{7.57 - 3(1.1667)^2\}}} = -0.97$$



Let $i=1,2,\dots,10$. x_i denote the reaction, measured in appropriate units, of the i th patient due to drug A. Let y_i denote their reaction to drug B. The observed values of the reactions are

i	x_i	y_i
1	1.9	0.7
2	0.8	-1.0
3	1.1	-0.2
4	0.1	-1.2
5	-0.1	-0.1
6	4.4	3.4
7	4.6	0.0
8	1.6	0.8
9	5.5	3.7
10	3.4	2.0



$$y = \beta_0 + \beta_1 x \quad \checkmark$$

$$Q = \sum_{i=1}^n \{y_i - (\beta_0 + \beta_1 x_i)\}^2$$

$$Q(\beta_0, \beta_1) \quad \frac{\partial Q}{\partial \beta_0} = 2 \sum_{i=1}^n \{y_i - (\beta_0 + \beta_1 x_i)\} (-1)$$

$$\begin{aligned} \frac{\partial Q}{\partial \beta_1} &= 2 \sum_{i=1}^n \{y_i - (\beta_0 + \beta_1 x_i)\} (-x_i) \\ &= -2 \sum_{i=1}^n x_i \{y_i - (\beta_0 + \beta_1 x_i)\} \end{aligned}$$

$$\frac{\partial Q}{\partial \beta_0} = 0 \Rightarrow \sum_{i=1}^n \{y_i - \beta_0 - \beta_1 x_i\} = 0$$

$$\Rightarrow \sum_{i=1}^n y_i - n\beta_0 - \beta_1 \sum_{i=1}^n x_i = 0 \quad \text{--- ①}$$

$$\frac{\partial Q}{\partial \beta_1} = 0 \Rightarrow \sum_{i=1}^n x_i \{y_i - \beta_0 - \beta_1 x_i\} = 0$$

$$\Rightarrow \sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0 \quad \text{--- ②}$$

$$n\beta_0 + \beta_1 \sum_{i=1}^n x_i - \sum_{i=1}^n y_i = 0$$

$$\beta_0 \sum x_i + \beta_1 \sum x_i^2 - \sum x_i y_i = 0$$

$$\beta_0 + \beta_1 \frac{1}{n} \sum x_i - \frac{1}{n} \sum y_i = 0$$

$$\text{or } (\beta_0 + \beta_1 \bar{x} - \bar{y} = 0)$$

$$\text{or } \boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$$

$$\frac{\beta_0}{1} = \frac{\beta_1}{n \sum x_i y_i - \sum x_i \sum y_i} = \frac{1}{n \sum x_i^2 - (\sum x_i)^2}$$

$$\Rightarrow \beta_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$\bar{y} = \beta_0 + \beta_1 \bar{x}$$

$$= \frac{n \sum x_i y_i - (n\bar{x})(n\bar{y})}{n \sum x_i^2 - n^2 \bar{x}^2}$$

$$\boxed{\hat{\beta}_1 = \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2}}$$

$$\boxed{\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x}$$

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

$$Q(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^n \{y_i - (\beta_0 + \beta_1 x_i + \beta_2 x_i^2)\}^2$$